Power Budget

Team Number: 107
Project Name: Simple Educational Robotic Arm - Arm Motion Control
Team Member Names: Yixiang Ren (Arm Motion Control subsystem)
Version: Rev C 2.3 (2026-11-06), matches schematic rev 2.3

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	H-bridge driver	FAN8100N	+1.8 - 9V	1	1500	1500	mA
	DC gearmotor 6V	DFRobot FIT0579	+6V	1	430	430	mA
	MCU board (Nano)	DM164150	USB +5V	1	500	500	mA
	Power LED (green)	WP710A10SGD	Vf 2.2V	1	25	25	mA
	Debug LED (yellow)	WP710A10YD	Vf 2.1V	1	20	20	mA

+6V Motor Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC gearmotor (stall)	DFRobot FIT0579	+6V	1	430	430	mA
	FAN8100N PVCC	FAN8100N	+1.8 - 9V	1	65	65	mA
	LM317 divider R11+R12	240R / 910R	+6V	1	5.2	5.2	mA
					Subtotal	500.2	mA
					Safety Margin	25%	
					Total Current Required on +6V Rail	625.25	mA
Regulator	+6V regulator (VR1)	LM317T	+3 - 40V in	1	1500	1500	mA
					Total Remaining Current Available on +6V Rail	874.75	mA

+5V Logic Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	FAN8100N VCC (logic)	FAN8100N	+2.2 - 9V	1	8	8	mA
	Power LED DS1 + R1	WP710A10SGD	+5V	1	2.8	2.8	mA
					Subtotal	10.8	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	13.5	mA
Regulator	+5V Regulator (VR2)	L7805CV	+7 - 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	1486.5	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 (allow.)	PIC18F57Q43	+1.8 - 5.5V	1	25	25	mA
	Debug LED DS2 + R10	WP710A10YD	+3.3V	1	3.6	3.6	mA
	Team-link outputs (x2)	330R + 10k	+3.3V	2	0.3	0.6	mA
	J4 3.3V pin (allow.)	J4 header	+3.3V	1	50	50	mA
					Subtotal	79.2	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	99	mA
Regulator	+3.3V LDO (on Nano)	MIC5353	+2.6 - 6V	1	500	500	mA
					Total Remaining Current Available on +3.3V Rail	401	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	9V wall adapter	GST25A09-P1J	90 - 264VAC	+9V	2550	2550	mA
Power Rails Connected to External Power Source 1	+6V rail via LM317T	LM317T	8.55V in	1	1500	625.25	mA
	+5V rail via L7805CV	L7805CV	8.55V in	1	1500	13.5	mA
	L7805CV quiescent	L7805CV	8.55V in	1	8	8	mA
					Total Remaining Current Available on External Power Source 1	1903.25	mA

External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	PC USB 2.0 port	USB 2.0	+5V	+5V	500	500	mA
Power Rails Connected to External Power Source 2	+3.3V rail via MIC5353	MIC5353	+5V in	1	500	99	mA
	Debugger (allow.)	nEDBG (on Nano)	+5V	1	100	100	mA
						0	mA
					Total Remaining Current Available on External Power Source 2	301	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	No battery	n/a	n/a		n/a	n/a	
					Battery Life	n/a	hours

Notes:

Supply voltage: LM317 needs about Vout + 2 V = 8 V; the 9 V adapter minus the 1N5819 drop (about 0.45 V) gives 8.55 V.
Source loads use each rail's required current (with margin), not the 1.5 A regulator ratings (3 A total would exceed 2.55 A).
Fuse check: current through F1 (1 A slow-blow) = +6V rail + +5V rail + L7805 quiescent = about 647 mA, below 1 A.
Thermal: VR1 dissipates (8.55 - 6) V x 0.63 A = 1.6 W worst case, so it uses heat sink 577102B00000G; VR2 is about 0.08 W.
Motor (DFRobot FIT0579) counted at stall (0.43 A); at its rated load it draws about 61 mA. 'allow.' = design allowance, not a datasheet va
Single unit (Arm Motion Control board). The Curiosity Nano (USB) and the 9 V supply grounds are tied together.